Mathematics for Computer Scientists COMP1046
Coursework 1

Deadline for submission: 23:59, 6 November 2025.
Submit as a PDF file via Moodle. Use student ID as your file name.
You may type your answer in Word or LATEX. A LATEX template is provided. Handwritten solutions are NOT accepted.
This coursework is out of 10 marks, with a penalty of 5% deducted for each day it is late.
Please work individually and show working process using the steps in the lectures.
Important: This coursework is designed to assess your understanding and application of core concepts in the module. Grading will place significant emphasis on your reasoning process and original analysis. While AI tools may be used as supplementary learning aids, their use must be properly cited, and the final submitted work must fundamentally represent your own original thought.

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1. **Given:**

$$\forall x \exists y (C(x) \to (D(y) \wedge L(y, x)))$$
$$\forall x \forall y (D(x) \wedge S(y) \to \neg L(x, y))$$

Prove: $\forall x (C(x) \to \neg S(x))$ using step-reason format. (2 marks)

2. Use mathematical induction to prove:

$$\forall n \in \mathbb{Z}, n \geq 1, \quad 7^n - 6n - 1 \equiv 0 \pmod{36}$$

(2 marks)

3. Let X be a non-empty set and let $\mathcal{F}$ be a non-empty family of subsets of X. That is, $\mathcal{F} \subseteq \mathcal{P}(X)$.

Prove the following set identity:

$$X \setminus \left(\bigcup \mathcal{F} \right) = \bigcap \{ X \setminus Y \mid Y \in \mathcal{F} \}$$

where,
$\bigcup \mathcal{F} = \{x \mid \exists Y \in \mathcal{F}, x \in Y\}$, in words, the union of $\mathcal{F}$ is the set of elements in at least one set of $\mathcal{F}$.

$\bigcap \mathcal{F} = \{x \mid \forall Y \in \mathcal{F}, x \in Y\}$, in words, the intersection of $\mathcal{F}$ is the set of elements in all sets of $\mathcal{F}$. (2 marks)

4. Given a set $X \neq \emptyset$. A relation R is defined on the power set $\mathcal{P}(X)$ of X as follows:

$$R = \{(A, B) \mid A, B \in \mathcal{P}(X),\ A \cap B \neq \emptyset\}$$

Determine whether this relation is reflexive, symmetric, antisymmetric, and transitive. Justify your answer. (2 marks)

5. You are a security engineer for a large tech company. Your network is protected by an AI-powered Intrusion Detection System (IDS). The system is designed to raise a high-priority alert if it detects patterns consistent with a sophisticated cyberattack. Based on industry threat intelligence, you initially estimate that on any given day, the probability of your network being under such a sophisticated attack is 2% (i.e., P(Attack) = 0.02). The IDS vendor provides the following performance metrics for their system:

   - True Positive Rate (Sensitivity): If an attack is occurring, the system will correctly raise a high-priority alert 98% of the time.
   - True Negative Rate (Specificity): If no attack is occurring, the system will correctly remain silent (no high-priority alert) 95% of the time.

   (a) One morning, you see that no high-priority alert has been raised. Given this information, what is the revised probability that your network is actually under an attack?

   (b) Your manager is concerned about false alarms. She asks you, "If the system does raise a high-priority alert tomorrow, what is the probability that it's a real attack and not a false alarm?"

   (c) Let the random variable X be the number of days, out of a 30-day month, that the IDS raises a high-priority alert. Assume the events on each day are independent. What is the expected value and variance of X?

   (d) A sophisticated cyberattack, if it occurs, causes an average of $RMB150,000$ in damages. Responding to a high-priority alert (whether it's a real attack or a false alarm) costs the company $RMB5,000$ in staff time and resources.

Let the random variable Y be the total cost incurred on a single day. What is the expected daily cost E[Y]?

(2 marks)